Are Generative AIs like Midjourney and Sora creative? The "Generative AI Mary's room" thought experiment.

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Are Generative AIs like Midjourney and Sora creative? The "Generative AI Mary's room" thought experiment.

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This article examines the debate of whether generative AI systems like Midjourney and Sora can be considered truly creative or are merely remixing existing data. It reviews Finke, Ward, and Smith's creative cognition approach which proposed constraints on human creativity. A new thought experiment, the "Generative AI Mary's Room", is proposed to explore whether a person with comprehensive artistic knowledge but no real-world experience could produce original creative works. The argument is that one who believes generative AI is not creative because it works based solely on previous artistic data, should not believe that Mary would be creative either, otherwise, they are inferring mysterious properties to the human mind.

Keywords: Generative AI, Machine Creativity, Midjourney, Sora, Creative Cognition.

Introduction

The rise of generative AI systems like Midjourney and Sora (Liu et al., 2024) has sparked a lively debate on what counts as creativity (Runco, 2023). It is possible to argue that these AI models are powerful creative tools that match human creativity by generating novel images and videos and other content and contend that generative models learn from existing data to identify patterns and create new permutations, similar to how human creators draw inspiration from past works. These models can produce novel artworks, images, videos and other content that appears highly creative to humans. Yet it is debatable whether such outputs should truly be considered creative expression.

It is possible to argue that even if generative AIs produce surprising or valuable outputs, they are ultimately constrained by simply recombining and remixing their training data in novel ways. AI models can learn to mimic and deviate from artistic styles but are still operating within bounded domains defined by their training distributions. Following this sort of argument, one could claim also that generative AI is an advanced plagiarism engine, repackaging and recontextualizing existing human creativity.

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What does it mean when something is novel?
adjective. of a new and unusual kind; different from anything seen or known before: a novel idea.

However, generative AI outputs do exhibit key hallmarks of creativity that make them more than mere remixed replications. State-of-the-art diffusion models can produce wildly imaginative "machine hallucinations" that diverge significantly from their training distributions through iterative feedback processes emulating aspects of the human creative cycle. The outputs have novelty, value, and meet traditional criteria for copyrightability despite their machine origins. But it is possible to view this as a form of high-tech plagiarism, where the AI repackages intellectual property without consent or compensation to the original creators. There can also be concerns about the potential of generative AI to displace human artists, writers, and coders by cheaply replicating their work (Avrahami, O., & Tamir, B., 2021).

This article focuses specifically on the argument that AI is not creative because it is merely recombining previous artwork. To address the issue, we discuss some limits of human creativity and propose a thought experiment called the Generative AI Mary's room.

What are some limits to human creativity?

The creative cognition approach, proposed by Finke, Ward, and Smith (1996), aims to demystify conventional ideas about creativity, challenging notions of "supernatural force" or "special individuals" being necessary for creative processes.

These researchers emphasize that creativity is an intrinsic aspect of human thought and should be investigated and understood in-depth. Their approach recognizes that everyone has the potential to be creative, provided they cultivate and develop cognitive skills and processes related to generating new ideas.

By studying creativity seriously yet without restricting its understanding, Finke, Ward, and Smith (1996) provide a more accessible and inclusive approach to exploring the creative phenomenon. They debunk myths and misconceptions, demonstrating that creativity is an intricate mental process influenced by contextual factors, prior knowledge, and external stimuli.

Creativity often results from complex internal mental work, where the interaction between knowledge, reasoning, and other more basic generative functions takes place. It emerges from this rich mental ecosystem, where associations, comparisons, and possibilities are constantly being processed.

Ronald A. Finke, Thomas B. Ward, and Steven M. Smith (1996) conducted a series of experiments probing the limits of human creativity, including the famous "alien creatures" study. Here are some relevant details on that work: In one of their key experiments, Finke, Ward, and Smith (1996) asked undergraduate students to draw beings from another planet that were unlike anything found on Earth. The students were told these alien creatures should not follow Earth's biological principles or laws of physics.

Despite the wide creative latitude given, the researchers found that the students' alien drawings overwhelmingly ended up resembling objects or life found on Earth, just with some minor tweaks or feature combinations. Common examples included bugs with humanoid bodies, snakes with arms and legs, or blobs with facial features.

The students appeared constrained by their direct experiences and struggled to transcend a blended, mutated version of terrestrial exemplars when attempting to conceive truly alien and divergent beings unprompted. Their creativity was limited by their own mental models shaped by their actual perceived realities. Even more interesting was the fact that there were not many differences from another group of students who were told to draw aliens that were *like* animals on earth.

This finding supported Finke, Ward, and Smith's (1996) hypothesis that creative cognition is often confined by mundane, traditional mindsets that impose inadvertent barriers to divergent exploration. Human creativity requires overcoming these cognitive constraints and restructuring conceptual landscapes to manifest original, transformative ideas completely distinct from familiar knowledge and experiences.

The researchers proposed methods like promoting mental shape-shifting and elaboration on initial generative concepts as techniques to expand creative boundaries. But their alien creatures study highlighted the challenge of bypassing conventional thought patterns to unlock radical, paradigm-breaking creativity unrestrained by tacit assumptions and terrestrial exemplars.

Finke, Ward, and Smith's (1996) work revealed the creative tension between combinatorial reorganization of familiar concepts versus truly disruptive, de novo novelty detached from any earthly constraints. It raised profound questions about the nature of creativity - whether it inevitably stems from recombining known elements, or if genuine originality arises disconnected from any experiential foundation. These issues resonate deeply with the philosophical debate over whether powerful AI models can exhibit true creativity independent of their training data.

Generative AI Mary's room

Mary's room is a philosophical thought experiment proposed by Frank Jackson (1985) as an argument against physicalism - the doctrine that the world is entirely physical with no non-physical elements like minds or consciousness existing separately from physical processes.

The thought experiment (Jackson, 1985) goes like this: Suppose Mary is a brilliant scientist who is forced to investigate the world from a black and white room via a computer screen. She specializes in the neurophysiology of vision and color, and she comes to know every physical fact about the processes in the brain involved in human color vision.

However, Mary has personally never experienced color herself, as she has been confined from birth to a room that is black and white. One day, Mary is freed from the room.

At this point the thought experiment asks: Will she learn anything new by leaving the room? Will she gain new knowledge of color qualia (the subjective, conscious experience of color) that factual knowledge alone could not provide?

If physicalism is true and all facts can be captured in physical terms, then it would seem that Mary already knows all there is to know about color vision from her complete neurophysiological knowledge learned in the room. But it is widely held that Mary will in fact gain a new perspective and learn something new - the first-person subjective experience of what it is like to consciously experience color qualia.

This intuition that Mary does learn something new forms the basis of the knowledge argument against physicalism. It suggests there are subjective, first-person experiential facts about consciousness that cannot be captured by third-person physical descriptions alone. Therefore, the argument claims, physicalism must be an incomplete theory of mind that fails to fully account for conscious experiences.

While an influential argument, responses include disputing that qualia are in fact non-physical, denying that Mary acquires new factual knowledge, or biting the bullet that complete physical knowledge would indeed allow deducing all experiences. The thought experiment continues to spur debate around whether consciousness can be explained in purely physical terms. There are many valuable responses to this argument (Dennett, 1991; Nemirow, 1990; Bigelow & Pargetter, 1990; Tye, 1986) and we are just going to take its structure as a thought experiment to think about creativity in AI.

So, the current version goes like so: Suppose Mary is a young girl who has been confined from birth to a room. However, this room is not black and white - rather, the walls are covered from floor to ceiling with artworks of every kind: paintings, drawings, sculptures, books, records, films, etc. This is the entirety of Mary's experience of the world.

Through careful study over many years, Mary masters an encyclopedic knowledge of all the artwork surrounding her. Through diligent study over many years, Mary gains comprehensive knowledge about the workings and patterns within this artistic data. She understands every style, genre, medium, technique, and cultural movement reflected in the works.

However, crucially, Mary has never directly perceived reality itself - no nature, people, objects, or first-hand experience outside of the artworks. Her only window into the world has been the creative representations and expressions of it contained in the art surrounding her vast room of artworks.

Mary herself has never actually created any original artwork. Her entire experience is confined to analytically studying and deeply comprehending the creative works of others lining her room's walls. She has never taken brush to canvas, pen to paper, or any other firsthand act of artistic expression herself.

The key question is: If Mary never leaves her room and experiences the world directly, instead remaining imprisoned but granted the ability to produce and output new artworks herself based solely on her analytical knowledge of the existing works, would Mary's creations be considered genuinely creative acts? Or would she merely be regurgitating remixed replicas derived from the data, without tapping into an experiential first-person spark of novel creative expression?

If one believes that what separates human creativity from that of AIs is that humans have experienced the world throughout years and not just a mass of paintings, then Mary's "creations" would be mere derivative works, reproductions lacking the essence of original creative expression.

However, if one thinks Mary would be creative but not the generative AI, then what is implied is that humans have a somewhat mystical creative ability that goes beyond perceiving content, reasoning about it, and generating interesting novelty.

The upshot, clearly, is that one who believes generative AI is not creative because it works based solely on previous artistic data, should not believe that Mary would be creative either, otherwise, they are inferring mysterious properties to the human mind.

Final Considerations

While there are certainly other important considerations around creativity of generative AI, such as the role of human prompting, human-in-the loop and so forth, this article purposely narrowed its scope to critically examine one of the core debates surrounding generative AI systems, whether their outputs should be considered truly novel creative works or are simply remixed, recombined versions of their training data.

It might also be important to note that there is another very important part of creativity which is left aside of both in this article and current AI, which is convergent thinking or exploratory phase reasoning (Cropley, 2006). Perhaps this sort of intricate reasoning is what people expect Mary to have and not the AI. This is currently a developing issue in AI (see Bellini-Leite, 2023). It would be interesting to know if people’s intuition would change after the development of a “system 2” for generative AI. That is, if artwork is created with a mix of “hallucination” on data and subsequent reasoning, planning, goals and motives for the artwork.

Ultimately, this work itself is simply a mixture of existing ideas from fields like creative cognition and philosophy of mind. The core concepts around creative constraints and combinatorial ideation are drawn from researchers like Finke, Ward, and Smith.

Philosophical considerations about the nature of creativity originate from thought experiments such as Mary's Room.

There is novelty in this piece but whether creativity is exhibited here in how these various perspectives are blended and reapplied is also questionable.

It may very well be that this work fails to transcend mere regurgitation of familiar ideas already traversed by others before. It could represent simply another iteration of humans' tendency to reflexively borrow and resequencing bits of their prior knowledge repositories, as exposed in Finke, Ward, and Smith's alien creatures study. A first-hand example of thought-patterns constrained by conceptual territoriality.

Then again, there is also a possibility that from the unique interconnections formed here, some semblance of novel insight has sparked into being. Perhaps this particular repackaging and mutated remixing of cognitive building blocks from creativity research and philosophical questions has converged into an expression distinctive enough to qualify as an original creative act, however incremental.

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